The Mountain Goats, Every Night

Scraping and modeling 34 years of live setlists from themountaingoats.fandom.com

The dataset

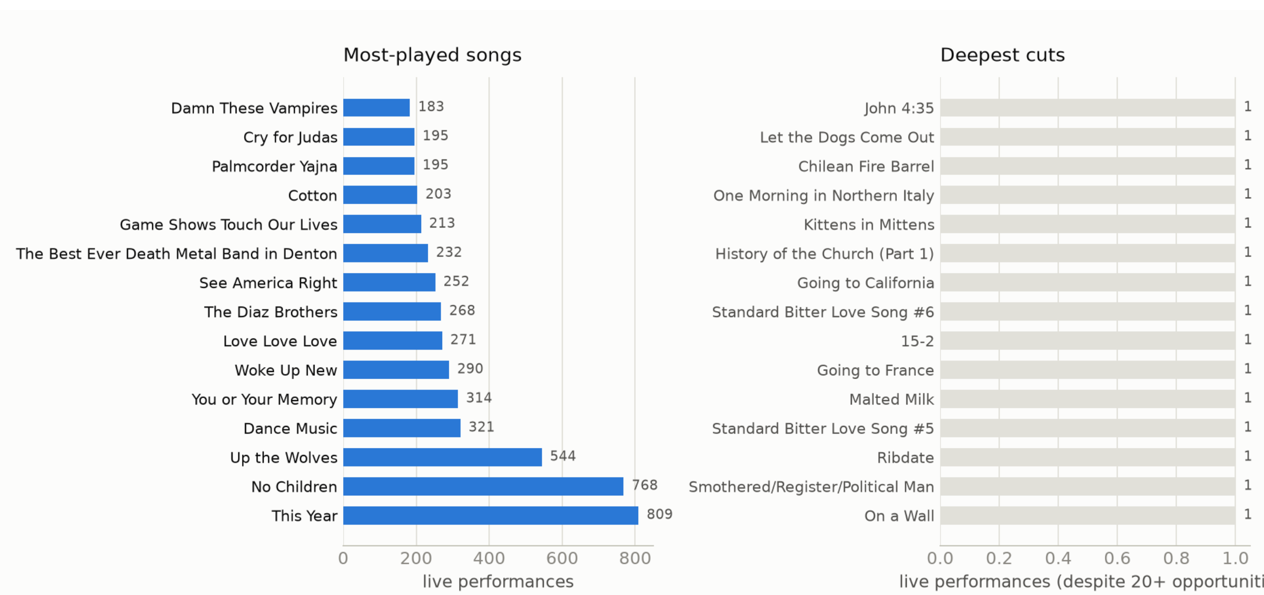
The wiki lists 1,579 live shows from 1992-05-31 through 2026-05-26 — solo John Darnielle sets, full-band tours, radio sessions, festival slots, and Extra Glenns/Extra Lens side-project shows. 1,317 of those have a documented setlist: 22,699 individual song performances across 845 distinct songs, spanning 97 named tours. 1,739 performances have at least one linked YouTube/Vimeo video.

Getting from a wiki category page to those numbers took a few real fixes:

- **Tour isn’t in the page text — it’s a wiki category** (e.g. *Peter Hughes Farewell Tour 2024*). Two earlier scraper attempts both missed this and scraped a null tour for every show.
- **Encores aren’t marked in the setlist table.** They’re stated in prose in the Notes section — “The encore was songs 19 through 23” — referencing the table’s printed order numbers. A small parser turns ~30 recurring phrasings of that sentence into a per-song encore number.
- **Song identity needed real normalization.** The wiki links each song to its own page, but the same song shows up under underscore-joined slugs, space-joined display text, and case variants of both. Left unmerged, this silently splits a song’s play count across rows — a real bug caught mid-project, where “Southwood Plantation Road” initially showed 1 play instead of its real count, 165.

The pipeline is two-stage: **fetch** downloads and locally caches every wiki page via the MediaWiki API (resumable, incremental on later runs), and **build** parses the cached HTML into tidy tables entirely offline.

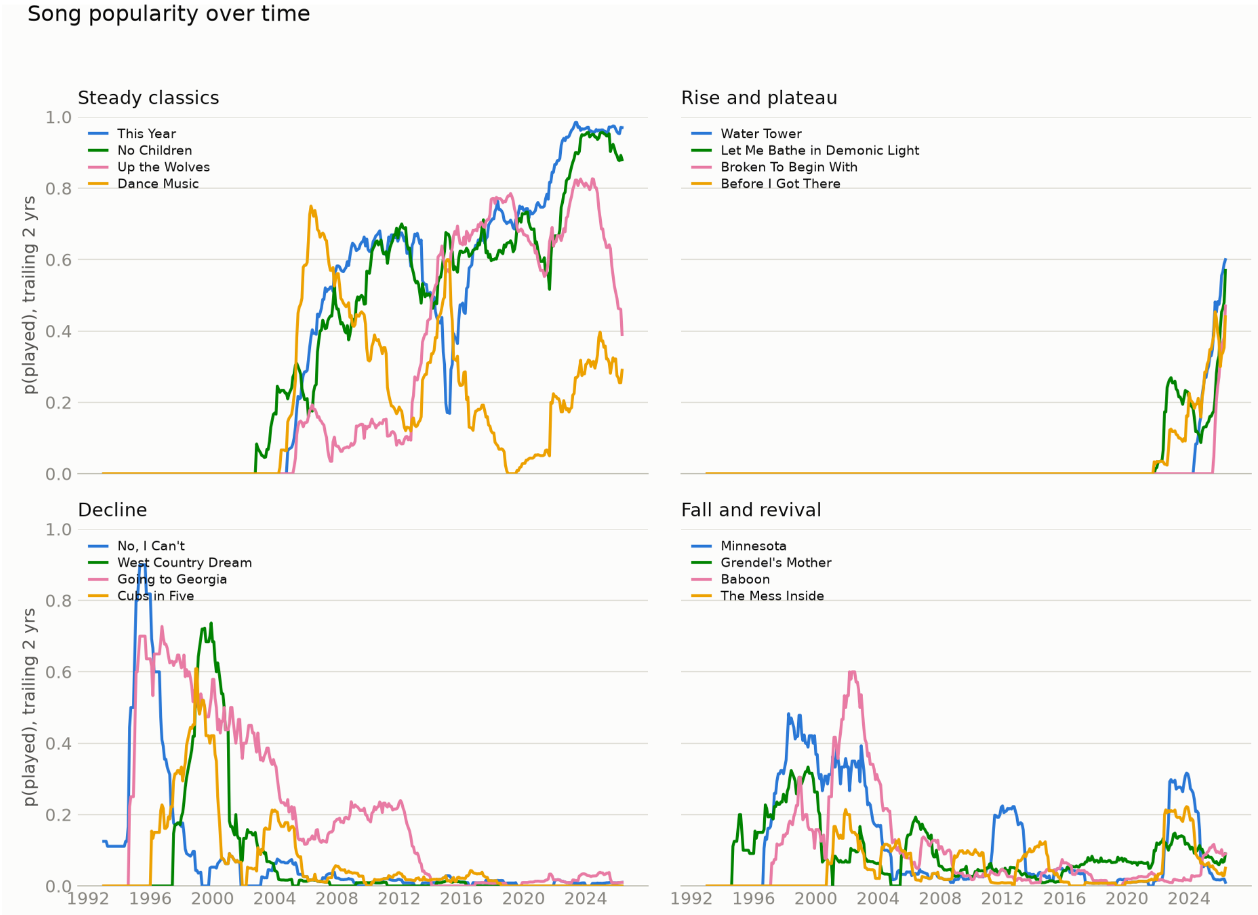
What gets played, and what doesn't



“This Year” and “No Children” anchor the top of the list — both played at a majority of all shows since entering rotation. The deep-cuts panel isn't just least-played-ever (most of the 845 songs have only 1–2 plays and aren't interesting on their own); it's the bottom of an opportunity-adjusted, shrinkage-smoothed play rate among songs with at least 20 chances to be played since their live debut — so a song only counts as a deep cut if it's been consistently passed over, not just recently written.

Song popularity over time

Treating each song's live history like a word's frequency in Google Ngrams — a trailing 2-year play rate sampled monthly — turns 34 years of setlists into trajectories: songs sit at zero before they're written, rise as they enter rotation, plateau, decline, and sometimes come back.



These four groups were picked automatically from the computed trajectories, not hand-selected: the most-played songs overall (steady classics); songs that debuted in the back half of the catalog's history and have climbed straight into rotation since (rise and plateau — a song appearing from nothing and taking hold); the biggest gap between early-career peak and recent play rate (decline); and songs with a real trough between two active periods (fall and revival).

The decline panel caught something the wiki itself corroborates: “Going to Georgia” was a fixture through the mid-2000s, then drops off hard — a 2012 show note has John Darnielle explaining, mid-show, that he considers it a “bullshit song” with an “asshole” narrator and doesn’t want to play it again. The data shows the falloff independent of that anecdote; the anecdote just explains it.

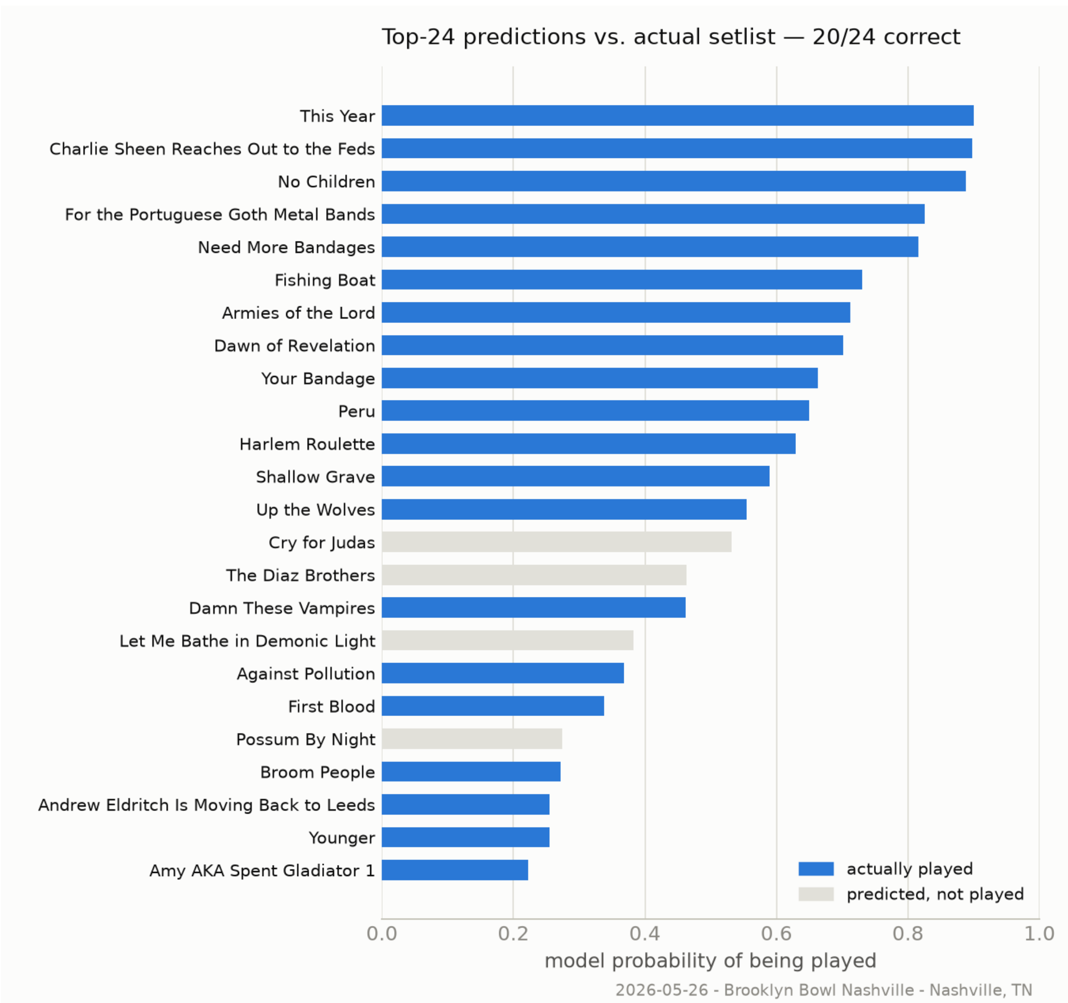
Predicting the setlist

Framed as one binary outcome per (show, song) pair — is this the night song X gets played? — every one of the 1,310 dated, setlisted shows becomes a training example, using only information available before that show: how recently and how often the song's been played, how it's fared on the current tour, its age, and whether it's a shortened-set special appearance (radio/festival/TV).

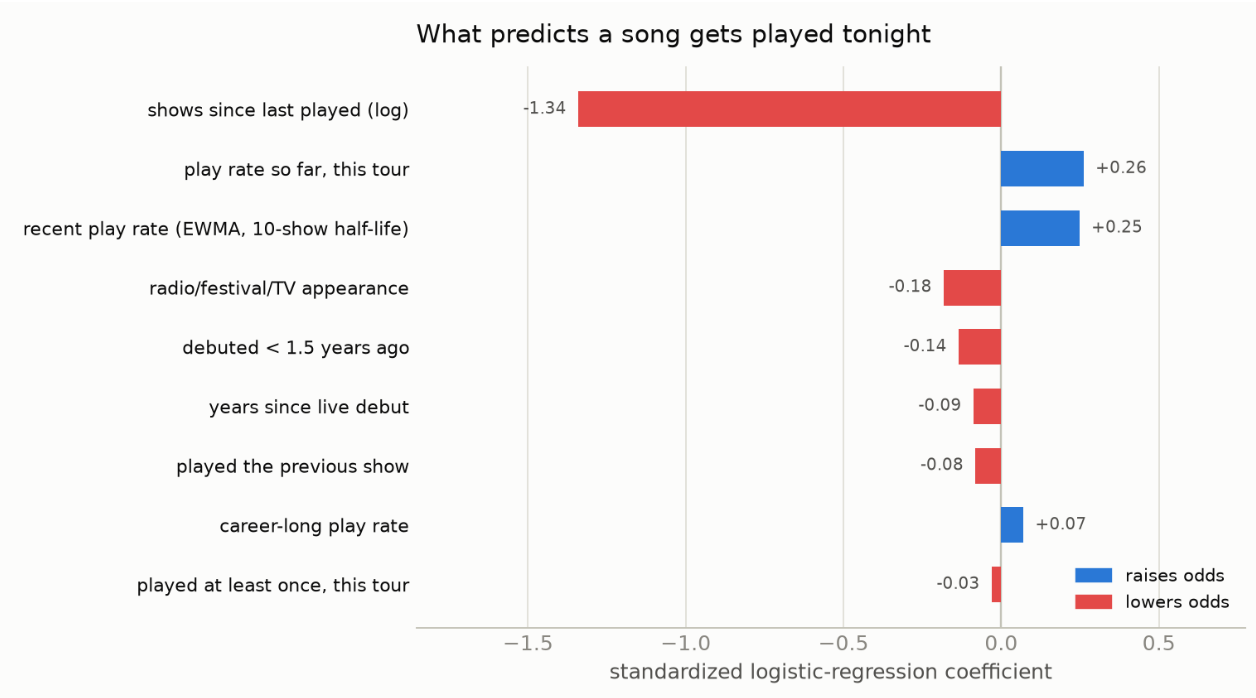
A logistic regression on those features is compared against a simple but strong baseline — an exponentially-decayed play rate alone — on a strict temporal split (train through 2022, test on the 207 shows from 2023 onward):

model	log-loss	Brier score	top-n setlist recovery
Baseline (decayed play rate)	0.0689	0.0162	51.0%
Logistic regression	**0.0563**	**0.0138**	**59.0%**

Top-n setlist recovery: for each show, take the model's n highest-probability songs, where n is the actual number played that night, and measure what fraction were right. On the most recent show in the data:



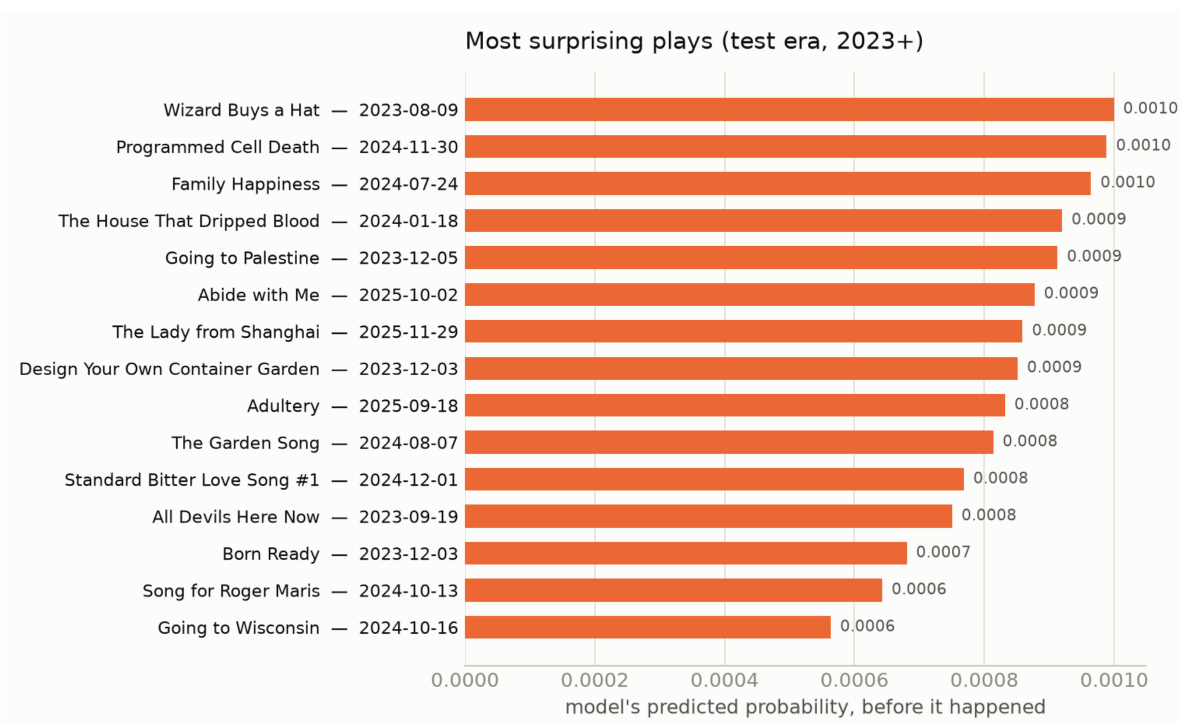
What actually predicts a setlist



Recency dominates everything else combined — if a song was played a few shows ago, it's overwhelmingly likely to come back soon. That's the visible mechanism behind the tour "rotation": John Darnielle appears to work from a live pool of songs and cycle through it, which is exactly what tour_rate (play rate so far, this tour) picking up an independent signal on top of recency confirms. Two smaller, genuinely interesting effects: conditional on recency, brand-new material is actually less sticky than catalog average, and radio/festival/TV appearances reliably favor hits over deep cuts.

When the model gets surprised

The flip side of a good model is a good list of its misses — nights the model gave a song almost no chance, and it got played anyway:



These are basically a machine-generated “rarities and one-offs” list: songs that came back from years of dormancy for a single night, often at a show with some specific occasion.

Caveats

- The wiki is fan-maintained and lists most, not all, shows — coverage is noticeably better from the mid-2000s on.
- The model has no access to anything outside setlist history — no lyrics, no album themes, no explicit “songs currently being toured.” tour_rate and song_age are proxies for that.
- Song identity is deduplicated via wiki link slugs where available; a handful of never-linked covers or rarities may still have unmerged spelling variants.

Reproducing this

```
PY=/Users/bdoughty/opt/miniconda3/envs/tmg-scrape/bin/python
$PY scrape.py fetch && $PY scrape.py build  # refresh the raw data
$PY analyze.py               # deep cuts, tour summaries
$PY predict.py               # the setlist model
$PY timeseries.py          # popularity-over-time series
$PY plot_report.py         # this report's figures
```

Full data dictionary in README.md; wiki-scraping gotchas and repo layout in CLAUDE.md.